

DAYANANDA SAGAR UNIVERSITY

Devarakaggalahalli, Harohalli, Kanakapura Road, Bengaluru South

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**SCHOOL OF
ENGINEERING**

Bachelor of Technology

in

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

A Project Report On

A Generative AI Framework for Automatic Comic Strip Generation

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(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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(2025 – 2026)

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CERTIFICATE

This is to certify that the project entitled “**A Generative AI Framework for Automatic Comic Strip Generation**” is carried out by **Akhila Rao D (ENG22AM0002)**, **Gaana Shree S (ENG22AM0014)** and **Nitya P Shetty (ENG22AM0037)**, bonafide students of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore, in partial fulfillment for the award of a degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025 - 2026**.

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DECLARATION

We, **Akhila Rao D (ENG22AM0002)**, **Gaana Shree S (ENG22AM0014)** and **Nitya P Shetty (ENG22AM0037)**, are students of the seventh semester B.Tech in Computer Science and Engineering (AI & ML) at the School of Engineering, Dayananda Sagar University. We hereby declare that the Minor Project titled “**A Generative AI Framework for Automatic Comic Strip Generation**” has been carried out by us and submitted in partial fulfillment for the award of a degree in **Bachelor of Technology in Computer Science and Engineering** during the academic year **2025–2026**.

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ACKNOWLEDGEMENT

It is a great pleasure for us to acknowledge the assistance and support of many individuals who have been responsible for the successful completion of this project work. First, we take this opportunity to express our sincere gratitude to School of Engineering & Technology, Dayananda Sagar University for providing us with a great opportunity to pursue our Bachelor's degree in this institution.

We would like to thank **Dr. Udaya Kumar Reddy K R, Dean, School of Engineering & Technology, Dayananda Sagar University** for his constant encouragement and expert advice.

It is a matter of immense pleasure to express our sincere thanks to **Dr. Jayavrinda Vrindavanam, Department Chairman, Computer Science and Engineering (Artificial Intelligence and Machine Learning), Dayananda Sagar University**, for providing right academic guidance that made our task possible.

We would like to thank our guide **Prof. Sahil Pocker, Assistant Professor, Computer Science and Engineering (Artificial Intelligence and Machine Learning)** for sparing his valuable time to extend help in every step of our project work, which paved the way for smooth progress and the fruitful culmination of the project.

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ABSTRACT

Comic creation traditionally involves multiple specialized skills including narrative writing, character design, visual illustration, and panel composition making the process labor-intensive and inaccessible for most users. This project presents a fully automated, AI-driven comic generation framework that integrates Large Language Models (LLMs), text-to-image diffusion models, and image post-processing modules to convert simple textual scenarios into complete comic strips. The system begins by interpreting a user-provided prompt and producing panel-wise narrative structures, character actions, dialogues, and scene descriptions using an LLM-based script generator. These structured outputs are then translated into visually coherent illustrations through a diffusion-based image synthesis engine optimized for comic-style rendering. To preserve consistency across panels, the pipeline incorporates prompt-refinement and controlled sampling strategies that maintain character identity, pose continuity, and stylistic uniformity.

Following image generation, the framework automatically renders speech text onto the panels using dynamic text placement and formatting logic, creating naturalistic speech bubbles and caption regions without manual design intervention. A layout engine combines individual panels into a complete horizontal comic strip, enabling seamless visual storytelling. The system also allows panel-level customization and regeneration to refine expressions, gestures, or dialogue while preserving continuity across the strip. By combining multimodal generative AI techniques with automated layout composition, this project democratizes comic creation, enabling non-artists, educators, storytellers, and hobbyists to rapidly produce expressive and visually appealing comic strips with minimal effort and no artistic experience.

Chapter 1

INTRODUCTION

Comics serve as a widely used medium for storytelling, visual communication, entertainment, and education. However, traditional comic production requires expertise in diverse creative domains, such as scriptwriting, illustration, character design, visual composition, and layout editing. This often discourages individuals without artistic or narrative skills from engaging in comic creation despite strong interest or imaginative ideas.

Recent advances in generative artificial intelligence particularly multimodal language models and high-fidelity diffusion-based image generators have created new opportunities to automate tasks that previously required human creativity. Large Language Models (LLMs) can interpret narrative prompts, generate coherent multi-step storylines, and structure them into panel-specific instructions. Meanwhile, diffusion models such as Stable Diffusion and its comic-optimized variants can synthesize high-quality illustrations that closely resemble hand-drawn comic art. Together, these models enable the development of an end-to-end system capable of generating both the semantic (story, script, dialogue) and visual (art, layout, bubbles) components of a comic strip.

This project introduces a unified system that integrates these AI capabilities into an automated comic generation pipeline. Users provide a simple text scenario ranging from a comedic situation to an adventure prompt and the system constructs a full comic sequence. The LLM interprets the scenario and produces panel-wise events, character actions, and dialogues. A diffusion model then transforms each panel description into stylized artwork. Finally, a rendering module positions speech text and merges the panels into a clean, coherent layout. By addressing challenges like character consistency, style alignment, and dialogue formatting, this system significantly reduces the complexity of comic creation and empowers users of all backgrounds to convert ideas into fully realized visual narratives.

Chapter 2

PROBLEM DEFINITION AND OBJECTIVE

2.1 Problem Definition

Creating comics is a multidisciplinary task that poses several challenges for non-experts:

- **Lack of Artistic Skills**

Users without drawing experience cannot confidently design characters, backgrounds, expressions, or action sequences.

- **Time-Consuming Workflow**

Traditional comic creation takes hours or days, making rapid prototyping nearly impossible.

- **Limited AI Tools Only Solve One Part**

Existing tools either generate images or assist in scripting but do not combine both into a coherent, complete comic strip.

2.2 Objectives

The primary objective of this project is to design and implement a fully automated, AI-driven framework capable of converting a simple textual scenario into a complete, stylized comic strip without requiring any artistic or scripting expertise from the user. The project strives to maintain stylistic continuity, character identity, and thematic coherence across panels, ensuring that the generated comic flows visually and narrative. Ultimately, the objective is to democratize the comic creation process by offering a user-friendly interface that enables anyone regardless of their artistic ability to develop expressive, high-quality comic strips with minimal effort.

2.3 Novelty of Proposed Approach

A key novelty of this project lies in its unified end-to-end pipeline that automates every stage of comic creation—from narrative generation to visual synthesis and final layout—using multimodal AI models. Unlike traditional creative tools that require users to manually write scripts, design illustrations, and arrange panels, this system leverages Large Language Models for structured panel scripting and diffusion models for producing stylized comic artwork directly from text descriptions. The integration of automatic text rendering and panel stitching further simplifies the workflow, enabling users to generate complete comic strips with minimal input.

The system offers a uniquely accessible and efficient platform for producing short-form comics without relying on artistic skill or complex editing software. It also introduces lightweight, modular, and API-driven architecture, which allows complex AI processes to be executed through a clean and minimal user interface. Rather than requiring users to manage model installations, GPU configurations, or local environments, the framework delegates script generation and image synthesis to cloud-based APIs. This design makes advanced generative AI accessible to users with standard hardware while maintaining scalability, reliability, and ease of deployment. The combination of LLM-driven storytelling and cloud-powered diffusion modeling, orchestrated through a simple web-based front end, offers a streamlined user experience that is uncommon in current comic creation or generative-art platforms.

This modularity also ensures that each component such as script generator, image generator, and layout engine can be independently improved or replaced, making the system adaptable for future advancements in AI creativity tools. The balance between automation and user-guided creativity makes the framework adaptable for a wide range of applications, including rapid prototyping for writers, personalized content generation for educators, and dynamic storytelling for digital media platforms. By transforming simple text prompts into expressive visual narratives, the system redefines the boundaries of digital art creation and demonstrates how generative AI can expand human creativity rather than replace it.

Chapter 3

LITERATURE SURVEY

The accelerated innovations in area of Artificial Intelligence has brought in new scope for projects related to generative AI. Combining techniques such as Natural language processing for processing the text and CNN for processing the image part has proven effective in enriching the model's capability to analyze and interpret the emotional, spatial and visual context of the given description. This analysis helps the stable diffusion model and DALL-E model in converting the text into visuals. Evaluation metrics such as accuracy, f1 Score, precession, and recall help in determining how relevant the generated image is to the given text.

The generation of comic strips can be enhanced by extracting semantic information from many datatypes, that is video , audio and text. A model was developed to extract video frames from a clip and group them into frames with dialogue and frames without dialogue using GIST similarity. The emotional information was derived from analysing the subtitles and audio sentiment which was used for generating the text balloons by feeding them into a SVM for selecting the shape and style of the balloon based on the detected emotion. The output panels were generated in manga like style using difference of Gaussian algorithm. These panels were then arranged into the given order using a data-driven stylistic layout model.

Studies tell that people who have the habit of reading comics are more empathetic towards the societal issues. A web application is more accessible to users than printed comics. This application generates comic strips based on the text entered by the user by classifying the text using fast text techniques and support vector machines and generating visuals using methods like generation of text bubbles, character recognition and voiceover interpretation. The deployment of sequential models contribute significantly for generation of comic strips. Based on the narration of the story which is considered as the criteria for the formation of comic strips can be performed using AI models. The layers of these models make use of decision making capabilities to transform theories into imageries making it more interactive.

A comparative study performed by comparing 15 AI comic strip generating platforms showed that integration of generative AI in generating comics results in efficient, faster and cost effective process where creativity meets automation. This study also showed a comparative analyses of various pricing of these platforms from freemium to paid subscription which concluded that free platforms attract greater number of users especially beginners. Another study performed on how reading comics benefits education showed that reading comics induces creative and artistic thinking in students. Diverse information which can be complex to explain can be effectively delivered to students by visual representations with descriptive texts as presented in a comic.

Chapter 4

METHODOLOGY

The development of the AI Comic Generator is centered around automating the creative workflow of comic strip production using modern generative AI models. The system is designed to interpret textual narratives, decompose them into comic panels, generate corresponding images, render dialogue, and combine the outputs into a coherent comic strip format. This methodology section explains the architecture, workflow, and design choices underlying the project, providing a comprehensive understanding of how the system operates end-to-end.

The pipeline begins with the Gradio frontend, which acts as the primary user interface. It receives a natural-language prompt from the user—typically a short story, description, or idea. The frontend performs basic validation, ensures that the input is safe and correctly formatted, and passes it to the backend for processing. Gradio was chosen due to its lightweight nature, real-time interactive capabilities, and seamless integration with Hugging Face Spaces, making the system highly accessible and easy to deploy.

Once the user input is captured, the next phase involves the OpenAI Large Language Model (LLM), responsible for narrative interpretation and panel breakdown. This model analyzes the prompt, extracts characters, actions, emotions, and environmental details, and converts the narrative into a set of structured panel descriptions. A strict JSON format is used for communication between the text and image generation components. The JSON includes fields such as number of panels, scene descriptions, character appearances, and dialogue text. Enforcing structured output ensures that the subsequent stages remain predictable, preventing formatting errors and enabling reliable automated processing.

Following JSON generation, the system applies a validation layer to check if the output adheres to the expected schema. Many real-world LLM outputs may contain trailing text, missing brackets, or deviations from the format. Error-handling strategies include regex cleanup, re-prompting the model, and providing fallback templates. This step enhances the robustness of the pipeline and prevents execution failures during image generation.

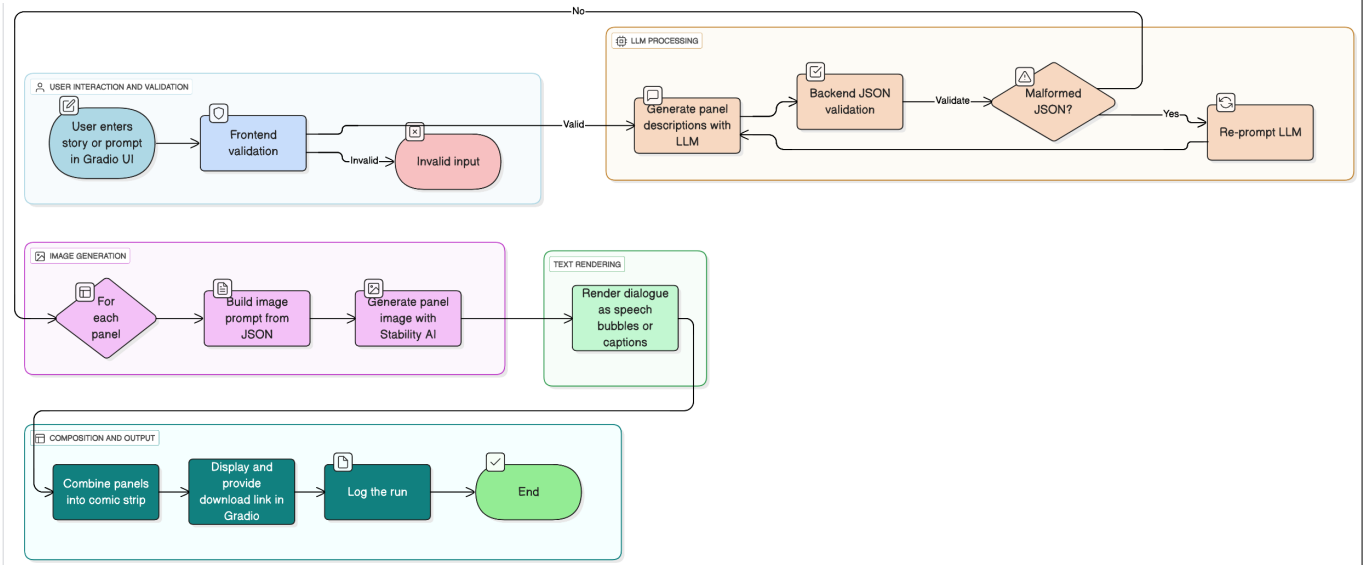


Figure 4.1: Workflow

With validated panel descriptions, the system moves on to visual generation using Stability AI’s diffusion model. For each panel, the text description is converted into an image prompt, enriched with style instructions such as “comic style,” “bold outlines,” or “cel-shaded coloring.” Specific diffusion parameters—such as seed, number of inference steps, cfg scale, sampler type, and image resolution (typically 512×512)—are applied to ensure visual consistency across panels. Safety filters and content-moderation responses from the API are also handled, enabling fallback prompts or style adjustments to ensure compliance with Stability’s generation standards.

After images are generated, the next stage involves text rendering and dialogue placement. Dialogue extracted by the LLM must be added to each panel as speech bubbles or captions. This is implemented using Python imaging tools such as PIL (Pillow). The system determines bubble shape, font selection, text wrapping, and position while ensuring readability and aesthetic coherence. A consistent visual language is maintained so that all panels feel part of the same comic narrative.

Once all images and dialogues are prepared, the system proceeds to comic strip composition. Panels are arranged into a single horizontal row, creating a traditional three- or four-panel comic layout. Spacing, margins, and alignment are carefully controlled. The final comic strip is rendered as a single combined image that the user can easily download or preview. This composition step ensures that even users unfamiliar with graphic design can produce polished, professional visuals effortlessly. To ensure reliability, the project incorporates multiple error-handling and robustness mechanisms. These include handling malformed JSON, retrying failed image generations, preventing API crashes caused by heavy inputs, and managing resource limits. Logging mechanisms record each step while excluding user content, ensuring both debuggability and privacy.

The methodology also includes considerations for security, ethical use, and resource management. API keys are stored securely using Hugging Face Secrets; user inputs are moderated to prevent unsafe content generation; and cost-aware decisions—such as generating small 512×512 images and caching repeated prompts—help keep the system efficient. These measures ensure that the tool remains safe, scalable, and affordable to operate.

Overall, the AI Comic Generator methodology showcases a complete generative-AI pipeline, from narrative interpretation to visual synthesis and composition. By combining LLMs, diffusion models, intelligent prompt engineering, structured data formats, and UI-driven deployment, the system provides an accessible, automated solution for transforming simple text prompts into high-quality comic strips. This approach not only democratizes visual storytelling but also demonstrates the integration of multi-modal AI technologies into a cohesive, user-friendly creative tool.

Chapter 5

REQUIREMENTS

5.1 Hardware Requirements:

- **Processor:** Dual-core processor (Intel Core i3 or AMD equivalent) capable of handling UI rendering and basic image composition tasks.
- **RAM:** Minimum 8GB to support Gradio interface execution, JSON processing, and lightweight image handling.
- **Storage:** At least 2–5GB of free space for storing project files, dependencies, temporary images, and logs.
- **Graphics Processing Unit (GPU):** Not required, as OpenAI and Stability AI handle all heavy model computation through cloud APIs.
- **Internet Connectivity:** Stable broadband connection required for API calls to OpenAI (text generation) and Stability AI (image generation).
- **Peripheral Devices:**
 - **Mouse and Keyboard:** Essential for interacting with the Gradio user interface and development environment.
 - **Monitor:** Standard single monitor is sufficient; larger screen preferred for reviewing generated comic panels.

5.2 Software Requirements:

- **Operating System:** Windows 10/11, macOS 11 or later, or Ubuntu 20.04+ compatible with Python 3.10 and required libraries.
- **Python Version:** Python 3.10 or higher to support API integrations, image processing, and the Gradio application.
- **Python Libraries:**
 - **gradio:** For building the user interface and hosting the comic generator.
 - **pillow:** For text rendering, speech bubble creation, and image composition.
 - **requests:** To communicate with OpenAI and Stability AI APIs.
 - **jsonschema:** For validating panel descriptions returned in JSON format.
 - **python-dotenv:** For securely loading API keys during development.
- **API Keys:** OpenAI API key (for panel text generation) and Stability AI API key (for image synthesis), stored securely using environment variables.
- **Code Editor:** Visual Studio Code, PyCharm, or any modern IDE for development, debugging, and script execution.
- **Version Control:** Git and GitHub for managing project files, version tracking, and deployment via Hugging Face Spaces.

Chapter 6

RESULT AND ANALYSIS

The AI Comic Generator is an advanced system designed to convert textual descriptions into conceptual comic-style images through a fully automated Machine Learning Operations (MLOps) pipeline. This pipeline ensures that the entire process, from input to output, is streamlined, reproducible, and scalable. The product leverages a multi-modal approach, where one model is dedicated to text generation and context understanding, ensuring that the narrative elements are coherent and aligned with user intent. Another model focuses on image generation, producing visuals that match the textual prompts, while a layout engine arranges these visuals into structured comic panels, maintaining the aesthetic and flow of a comic strip.

Although the system integrates two separate APIs to handle different models, it has been observed to operate smoothly even in CPU-based environments, which demonstrates its efficiency and optimization. This is particularly significant because many AI image generation systems require GPU acceleration, whereas this product achieves reliable performance without such hardware, making it more cost-effective and accessible.

Performance testing revealed predictable scaling behavior. On average, the generator requires 10 to 13 seconds to produce a panel of three images, while generating six images in a panel takes about 18 to 22 seconds. These results highlight that the time taken increases proportionally with the number of panels, which indicates stable scalability. The near real-time generation speed ensures that users can interact with the system without long delays, making it suitable for creative workflows, prototyping, and interactive applications.

Overall, the AI Comic Generator demonstrates robustness, adaptability, and efficiency. Its ability to integrate text understanding, image generation, and layout design into a single automated pipeline makes it a production-ready solution. By balancing speed, quality, and user experience, it provides a powerful tool for artists, storytellers, and enterprises looking to create comic-style visuals quickly and effectively.

Chapter 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project demonstrates the potential of multimodal generative AI to fully automate the traditionally complex and skill-intensive process of comic creation. By integrating Large Language Models for narrative generation and diffusion-based image models for visual synthesis, the system successfully converts simple text scenarios into coherent, visually appealing multi-panel comic strips. The automated pipeline for script generation, image synthesis, dialogue rendering, and panel layout provides a seamless user experience that eliminates the need for artistic expertise, manual design effort, or specialized software tools. The framework not only accelerates the comic creation workflow but also democratizes visual storytelling by making it accessible to a wide audience, including students, educators, hobbyists, and creative professionals. Overall, the project highlights how AI can enhance creative expression, streamline content generation, and open new pathways for interactive digital storytelling.

7.2 Future Scope

The system offers several promising directions for future enhancement. One major extension involves expanding stylistic control, allowing users to customize comic styles such as manga, chibi, noir, watercolor, pixel-art, or specific artist-inspired aesthetics. Incorporating more advanced speech-bubble generation models such as AI-driven bubble shape prediction.

From a broader perspective, the framework can evolve into a fully featured content creation suite that supports animated panels, motion comics, or even short AI-generated videos. Additionally, future versions can support dynamic panel layouts, multi-page comic books, and interactive editing tools where users can refine individual panels, modify character expressions, or regenerate specific scenes while preserving continuity.

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